MILITARY SPECIFICATION

EPINEPHRINE SUSPENSION, STERILE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Sterile Epinephrine Suspension (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

STANDARDS

FEDERAL

FED-STD-142 - Parenteral Preparations.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6505

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE, FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Applications for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be Sterile Epinephrine Suspension and shall be in accordance with all applicable requirements of FED-STD-142, and as specified herein.

S2. Classification. Shall be type I, class 2. Shall be suitable for subcutaneous use.

S5.2.2 The following is added to this paragraph:

Shall be a 1:200 sterile suspension of epinephrine suitable for subcutaneous use.
Each 5 ml immediate container of sterile epinephrine suspension shall contain a quantity of Epinephrine, USP, equivalent to 5 mg per ml within the applicable assay limits.

S6.1.2.1 Vehicle shall be water for injection.

3.1.1 Assay - total epinephrine. Each milliliter of sterile epinephrine suspension shall contain not less than 4.5 mg and not more than 6.0 mg equivalent to not less than 90 percent and not more than 120 percent of the labeled amount of epinephrine when tested as specified in 4.5.1.

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3.1.2 Assay - epinephrine suspended phase. Each vial of sterile epinephrine suspension shall contain in suspension not less than 60 percent and not more than 85 percent of the total epinephrine content when tested as specified in 4.5.2.

3.1.3 pH. pH of epinephrine suspension shall be not less than 7.5 and not more than 8.1 when tested as specified in 4.5.3.

3.2 Ingredients.

3.2.1 Epinephrine. The epinephrine used in the manufacture of the suspension shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.3 Other ingredients. All other ingredients used in the manufacture of the the suspension shall comply with S5.1 of FED-STD-142.

3.4 Expiration dating period. The expiration dating period shall be not less than 21 months.

3.4.1 Delivery. Not more than 3 months of the expiration dating period shall have elapsed at the time of delivery to the Government.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Sampling. Sampling shall be conducted in accordance with FED-STD-142.

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4.3 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.5 Tests. Tests shall be performed to determine compliance with all requirements of this specification.

4.5.1 Assay - total epinephrine.

4.5.1.1 Preparation of reagents.

Buffer solution. Mix 42 gram of sodium bicarbonate and 50 gram of potassium bicarbonate with 180 ml of purified water. Mix 37.5 gram of aminoacetic acid and 17 ml of stronger ammonia water with 180 ml of purified water. Mix the two solutio add purified water to bring the volume to 500 ml and mix until solution is complete.
Ferro-citrate solution. On the day needed, dissolve 250 mg of crystalline ferrous sulfate and 25 gram of sodium citrate in sufficient purified water to make 500 ml of solution.

4.5.1.2 Preparation of standard solution. Accurately weigh about 40 mg of Epinephrine Bitartrate, USP Reference Standard, previously dried in a vacuum desiccator over silica gel for 18 hours. Transfer to a 100 ml volumetric flask containing 400 mg of sodium bisulfite dissolved in 50 ml of purified water. Bring to the mark with purified water and mix well.

4.5.1.3 Preparation of sample solution. Remove all seals and stoppers from 3 vials of the sterile epinephrine suspension, and add 1 drop of hydrochloric acid to each vial. Replace the rubber stoppers in their respective vials and gently mix the contents until the suspended material has dissolved. Pool the contents of the vials in a 30 ml beaker, mix well and quantitatively transfer 10.0 ml of the solution to a 200 ml volumetric flask, bring to the mark with purified water and mix well.

4.5.1.4 Procedure.

Transfer 10.0 ml of the epinephrine bitartrate standard solution to a 100 ml volumetric flask labeled "standard". To each of eight 100 ml volumetric flasks, labeled in sequence 1 through 8, transfer 10.0 ml of the final dilution of sample prepratioon. To the flask containing the standard, add 10.0 ml of ferro-citrate solution. By means of a 50 ml burette, add to each flask in the sample series the volume of ferro-citrate solution as indicated in the following chart:

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Flask No.	1	2	3	4	5	6	7	8
ml Ferro-citrate solution	1.0	2.0	3.0	6.0	8.0	10.0	14.0	16.0

Add to each flask, standard and samples, 5 ml of buffer solution, mix gently and allow to stand in subdued light for 20 minutes. Bring to the mark with purified water and mix well. Determine the absorbance of each solution in 1 cm matched cells with a suitable Spectrophotometer at 530 nm (± 2 nm) using purified water as the blank. Using regular coordinate paper, plot the absorbances of each of the sample solutions (flasks 1 through 8) vs the corresponding volume of ferro-citrate solution added. Draw the best straight line through the points obtained from flasks 1 through 3 (A); draw the best straight line through the points obtained from flasks 4 through 6 (B); draw a straight line through the points obtained from flasks 7 and 8 (C). Extrapolate lines (A) and (B) to a point of intersection. The value of the absorbance at this intersection (Au) is the absorbance of the epinephrine present in the sample. Line (C) should be a plateau.

Calculation:

$$\frac{Au \times Ws \times 10 \times 200 \times 100 \times f}{As \times 100 \times 100 \times 10 \times 10} = \text{mg total epinephrine/ml Sterile Epinephrine Suspension}$$

where

Au = absorbance at 530 nm of sample solution as determined from the point of intersection of the lines (A) and (B).
As = absorbance of the standard epinephrine bitartrate solution final dilution at 530 nm.
Ws = mg weight of epinephrine bitartrate Reference Standard taken.
f = 0.5497.

4.5.2 Assay - Epinephrine suspended phase.

4.5.2.1 Preparation of reagents.

Buffer solution. Mix 42 gram of sodium bicarbonate and 50 gram of potassium bicarbonate with 180 ml of purified water. Mix 37.5 gram of aminoacetic acid and 17 ml of stronger ammonia water with 180 ml of purified water. Mix the two solutions, add purified water to bring the volume to 500 ml and mix until solution is complete.
Ferro-citrate solution. On the date needed, dissolve 750 mg of crystalline ferrous sulfate, 500 mg of sodium bisulfite, 0.5 ml of 1 N hydrochloric acid and 5.0 gram of sodium citrate in sufficient purified water to make 100 ml of solution.
Sodium bisulfite solution. (1:500).

4.5.2.2 Preparation of standard solution.

Accurately weigh about 40 mg of Epinephrine Bitartrate USP Reference Standard, previously dried in a vacuum desiccator over silica gel for 18 hours. Transfer to a 100 ml volumetric flask containing 400 mg of sodium bisulfite dissolved in 50 ml of purified water. Bring to the mark with purified water and mix well.

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4.5.2.3 Preparation of sample solution.

Remove the outer aluminum seal and inner cap from a vial of sterile epinephrine suspension and gently agitate the contents to obtain a uniform suspension. Invert the vial, puncture the rubber cap using a 1-1/2 inch, 21 gauge needle on a 5 ml clean, dry hypodermic syringe and remove the entire contents. Adjust the contents of the syringe to exactly 5.0 ml making certain that no air bubbles are present. Without delay expell the contents of the syringe into a 15 ml sintered glass filter of medium porosity, mounted in a suitable Filtrator. Collect the filtrate in a 250 ml beaker and discard. Replace the beaker with a 100 ml volumetric flask, add 3 ml of purified water and 1 drop of hydrochloric acid to the residue in the filter. Mix the acid solution with the residue using a small glass rod and with the aid of suction, filter this solution directly into a 100 ml volumetric flask. Wash the filter with repeated 10 ml portions of sodium bisulfite solution (1:500) until about 80 ml of filtrate have been collected. Remove the flask from the filtrator, bring to the mark with sodium bisulfite solution (1:500) and mix well.

4.5.2.4 Procedure.

Transfer 10.0 ml of the standard solution and 10.0 ml of the sample solution to separate 100 ml volumetric flasks. Add to each flask about 70 ml of sodium bisulfite solution (1:500), mix well, then add to each flask 1.0 ml of ferro-citrate solution and 10 ml of buffer solution. Bring to the mark with sodium bisulfite solution (1:500) and mix well. Allow to stand for 30 minutes, then determine the absorbance of the standard and sample solutions in 1 cm matched cells with a suitable Spectrophotometer at 530 nm ($\pm$ 2 nm), using purified water as the blank.

Calculation:

$$\frac{Au \times Ws \times 10 \times 100 \times 100 \times f}{As \times 100 \times 100 \times 10 \times 5} = \text{mg suspended epinephrine/ml Sterile Epinephrine Suspension}$$

where

Au = absorbance of sample solution 530 nm
As = absorbance of standard epinephrine bitartrate solution at 530 nm
Ws = mg weight of epinephrine bitartrate Reference Standard taken
f = 0.5497

$$\frac{A \times 100}{B} = \text{Percent of total as suspended epinephrine}$$

where

A = mg suspended epinephrine/ml sterile epinephrine suspension
B = mg total epinephrine/ml sterile epinephrine suspension

4.5.3 pH. pH shall be determined using the USP method except test shall be conducted at room temperature (23 $\pm$ 2° C).

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186 and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.2.1 Unit of issue. One bottle, as specified, constitutes one unit of issue.

5.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available vial considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. At the option of the contractor, each unit may be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate package. Twelve (12) units of issue shall be packaged as specified in 5.2.7.1 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. Five hundred seventy-six (576) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Five hundred seventy-six (576) units of issue shall be packed as specified in 5.4.3.1.1 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below.

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the expiration date.

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5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the National Stock Number.

NOTES: (1) The National Stock Number shall not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

5.4.2 Intermediate package. Intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. Type I shelf life markings as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° C (59° - 86° F)".

5.4.3 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° C (59° - 86° F)".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following item:

National Stock Number	Item Identification
6505-01-022-2402	EPINEPHRINE SUSPENSION, STERILE, 1:200, 5 mg per ml, 5 ml

Preparing activity:
DLA-DM

Project No. 6505-1987

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-E-37840(DLA-DM) EPINEPHRINE SUSPENSION, STERILE

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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